Assignment - 1.5

Day 5 : **Assemble and Disassemble a PC Practical :** Full Hands on session : Assemble and Disassemble a desktop PC Task : Document the process with labeled steps.

Task – 1

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

List of the all-main components of my Desktop PC after opening the cabinet :-

1. **Motherboard:-**
   The main circuit board of the PC. It connects and allows communication between the CPU, RAM, storage, GPU and other components.

2. **CPU(Processor) :-**
   The main processing unit of the computer. It executes instructions and performs calculations required to run programs.

3. **RAM :-**
   Temporary memory used to store data and programs that are currently being used. More RAM helps the computer handle multiple tasks smoothly.

4. **GPU (Graphics Processor) :-**
   Processes graphics, images, and videos. It can be integrated into the CPU or provided as a dedicated graphics chip.

5. **SSD / Storage :-**
   Permanently stores the operating system, applications, and personal files. SSDs provide fast data access.

6. **Battery :-**
   Supplies power to the laptop when it is not connected to an electrical outlet.

7. **Cooling Fan :-**
   Moves air through the cooling system and helps remove heat generated by the CPU and other components.

8. **Heatsink :-**
   Transfers and dissipates heat away from the CPU/GPU to keep them at a safe temperature.

9. **Wi-Fi / Bluetooth Card :-**
   Provides wireless network and Bluetooth connectivity for internet access and connecting wireless devices.

10. **CMOS Battery :-**
    Maintains the laptop's BIOS/UEFI settings and system clock when the main battery is disconnected.

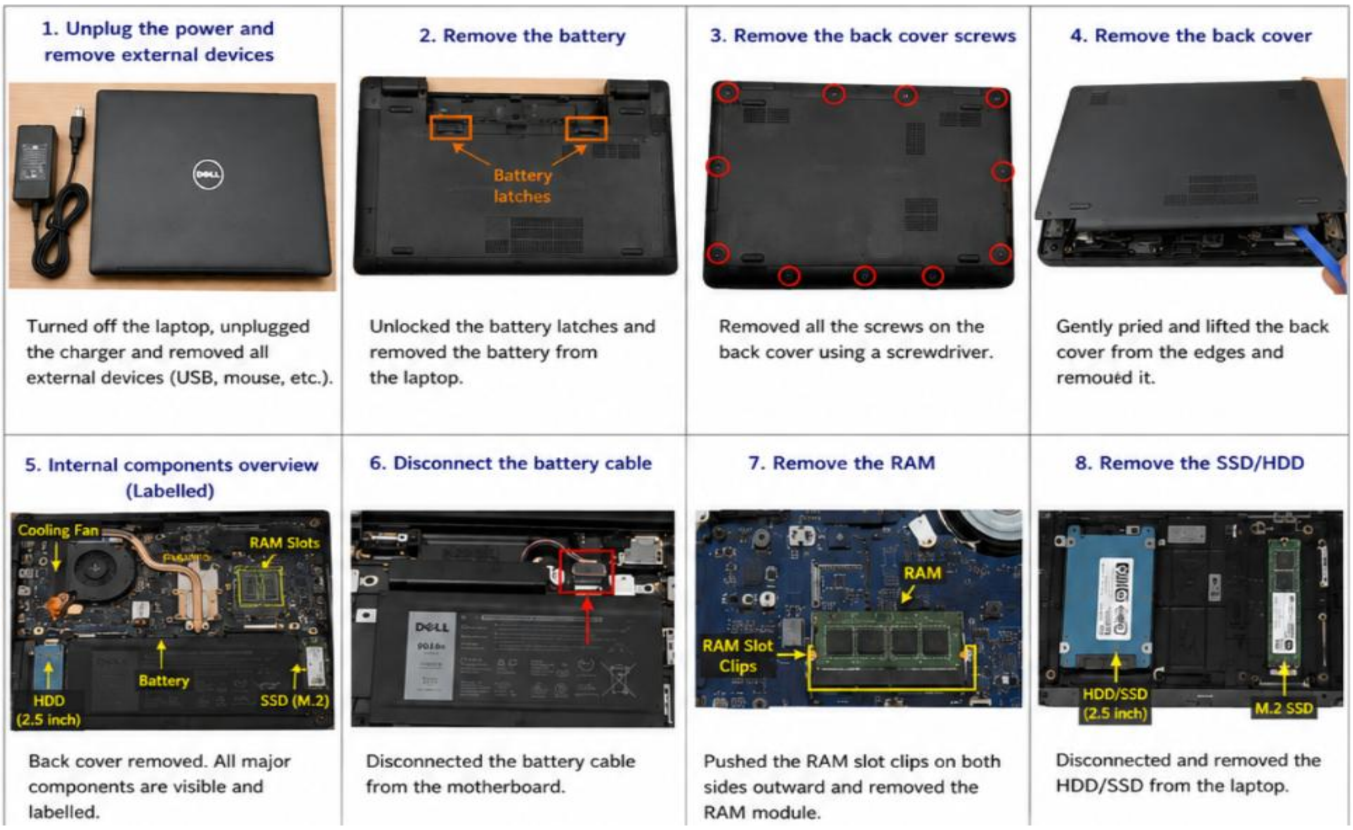
11. **Speakers :-**
    Convert electrical signals into sound for music, videos, notifications, and calls.

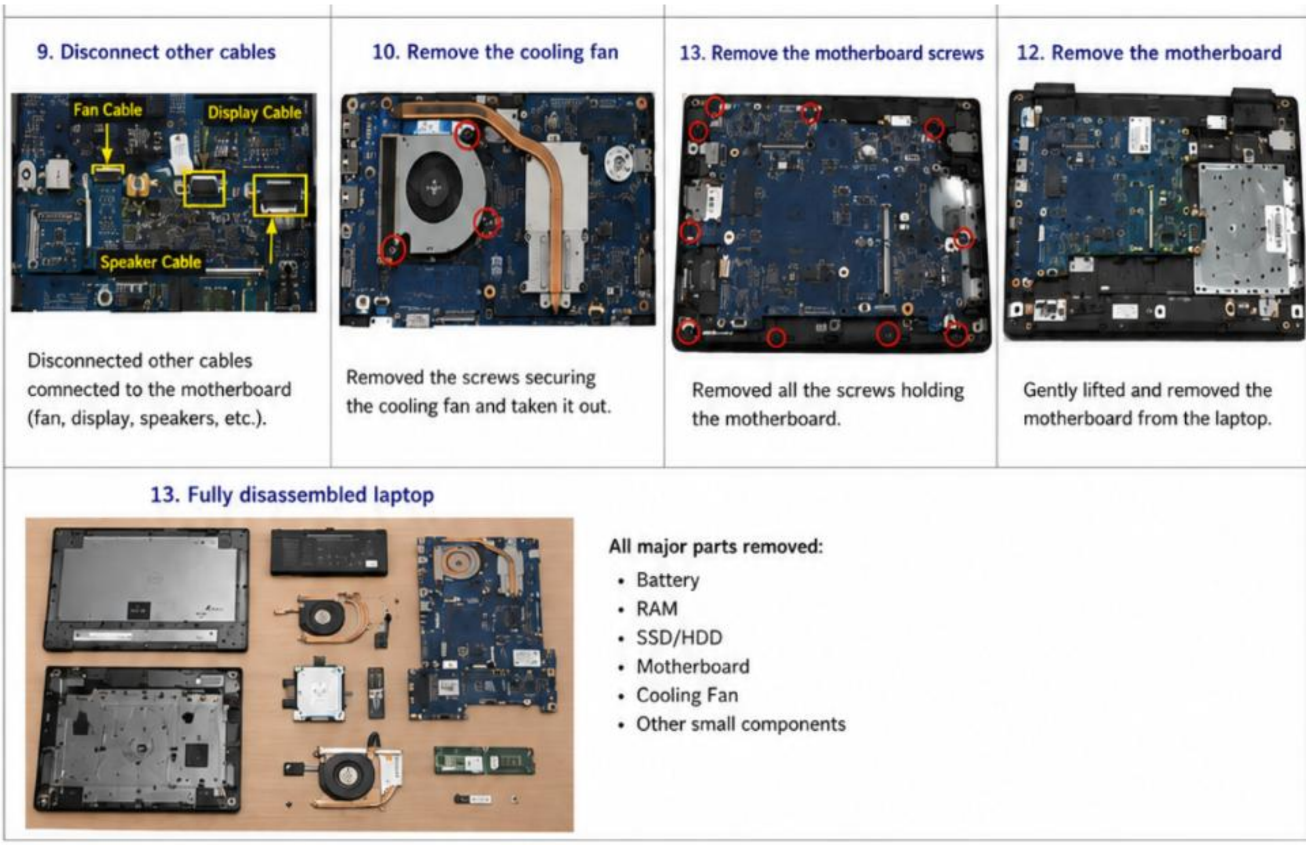
12. **Ports & Connectors :-**
    Provide connections for USB devices, headphones, charging, displays, and other peripherals.

Task – 2

Take clear photos or draw Labelled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

First of all, I cannot disassemble my laptop by myself, so I had these images created from AI and then studied them to understand the process.





Now , The PC is fully disassemble.

Task – 3

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

1. Install the motherboard

Placed the motherboard in the correct position and aligned all the ports with the slots on the chassis. This is the first step because all other components connect to the motherboard.

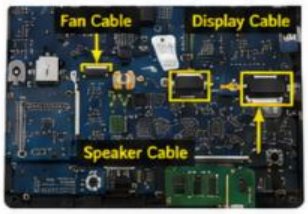
2. Secure the motherboard screws

Tightened all the screws to secure the motherboard firmly. This keeps the board stable and prevents movement or damage during use.

3. Install the cooling fan

Placed the cooling fan with the heat sink back in position and screwed it properly. This ensures proper heat dissipation and prevents overheating.

4. Connect other internal cables



Connected internal cables (fan, display, speaker, etc.) to their respective ports. These cables are essential for device functionality.

5. Install the SSD/HDD

Inserted the SSD/HDD into its slot and secured it. Storage is installed before closing the system so that it is ready for the operating system.

6. Install the RAM

Inserted the RAM module into the slot at an angle and pressed down until the clips locked into place. RAM is required for the system to boot and run.

7. Connect the battery cable

Connected the battery cable to the motherboard. This restores power to the system.

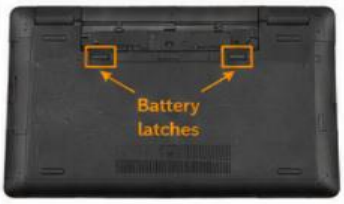
8. Place the back cover

Placed the back cover in position and aligned the edges properly. This protects internal components and ensures a proper fit.

9. Secure the back cover screws

Tightened all the screws to secure the back cover. This keeps the laptop safe and ensures proper cooling and structure.

10. Install the battery



Inserted the battery into the compartment and locked the latches. The battery powers the laptop when unplugged.

11. Connect external devices

Connected the charger and any other external devices that were removed.

12. Power on the laptop

Pressed the power button to turn on the laptop and verified that it boots correctly.

14. Reassembly Complete

The laptop has been reassembled successfully and is ready to use.

Task – 4

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Checklist of Safety Precautions are given below :-

- **Switch Off the Laptop** :- Prevents electrical accidents and protects internal components from damage.
- **Disconnect the Charger :-** Ensures that no external power is supplied while opening the laptop.
- **Remove/Disconnect the Battery :-** Prevents accidental power flow to the motherboard during disassembly.
- **Work on a Clean and Dry Surface :-** Prevents dust, moisture and small particles from entering the laptop.
- **Do Not Use Excessive Force :-** Prevents breaking clips, connectors, cables or the laptop casing.
- **Follow the Dell Service Manual/Guidelines :-** Model-specific instructions help ensure that components are removed and installed correctly.

Required Tools are given below :-

Tool	**Purpose**
Plastic Pry Tool	Used to open the back cover without damaging the plastic casing.
Tweezers	Useful for handling small connectors, screws and cables.
Flashlight	Helps to clearly see small screws, cables and connectors inside the laptop.
Soft cloth / mat	Protects the laptop and components from scratches and accidental damage.
Clean workspace	Provides enough space to safely place components and screws.
Small Parts Container/Magnetic Tray	Keeps screws and other small parts organized.

Conclusion :-

Following proper safety precautions and using the correct tools is important during laptop assembly and disassembly. It reduces the risk of electrical damage, ESD damage, broken connectors and lost screws. Proper preparation also makes the process safer, easier and more organized.